

BRICS • SUSTAINABILITY

VayuSetu

Federated, hyperlocal air-pollution intelligence for BRICS capitals

Citizen photos, ground stations and satellites fused into verified street-level hotspots, and routed to the authority that can clear them.

Gemini

Earth Engine

BigQuery ML

Firebase

Maps Platform

THE PROBLEM

A city-wide AQI hides the air people actually breathe



Hyperlocal sources go unseen

Garbage fires, construction dust and industrial stacks sit kilometres from the nearest monitor, so they vanish into the city average.



Pollution crosses borders

Crop-residue burning and trans-boundary haze drift across state and national lines faster than any single agency can track.



Data rarely reaches a crew

Dashboards publish numbers, but no one is told which street to inspect or which department owns the fix.



One monitor, many invisible sources, and smoke that does not stop at the border.

OUR SOLUTION

From a citizen's photo to a dispatched crew



01 SENSE

Citizens report with a photo, a voice note in their own language and a pin. Stations and satellites watch continuously.



02 VERIFY

Gemini vision confirms smoke, dust, fire or haze, and flags screenshots, stock images and duplicate photos.



03 FUSE

Each report is scored against the nearest station, Sentinel-5P NO₂ and aerosol, NASA FIRMS fires and live wind.



04 ACT

Verified hotspots are ranked and dispatched, with Gemini-drafted work orders and WhatsApp updates to reporters.

4

independent evidence streams

~0.7 km²

H3 hotspot cells, street scale

6

evidence-based promotion tiers

13

interface languages

BUILT FOR BRICS

Seven capitals, one shared engine



Highlighted: BRICS member states. Lines: the federated network of live capitals.

7

BRICS capitals live: New Delhi, Beijing, Moscow, Pretoria, Abu Dhabi, Jakarta, Brasília

7

national air-quality standards applied, each country's own legal limits

1

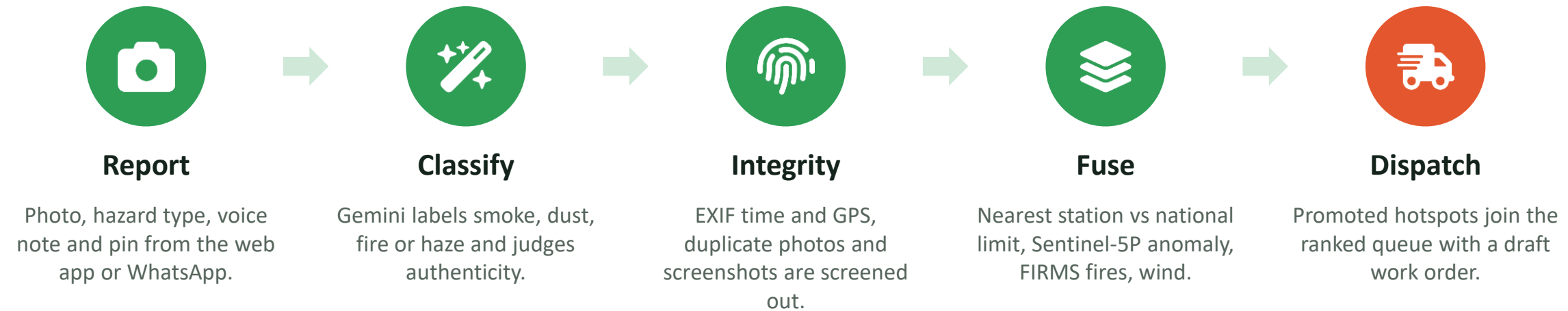
common data model, H3 grid and API across every city

WHAT MAKES IT DIFFERENT

Not another AQI dashboard

	Typical AQI dashboards	VayuSetu
Spatial scale	One city-wide average from a few monitors	Street-level H3 cells of ~0.7 km ²
Evidence	A single sensor feed	Citizen photos + ground stations + Sentinel-5P + NASA FIRMS + wind, fused
Trust	Reports taken at face value	Gemini vision, EXIF, duplicate and screenshot checks; promotion needs independent agreement
Action	Displays a number	Ranked queue, dispatch and resolve; work orders routed to the responsible agency
Cross-border	Stops at the city limit	Upwind attribution with live wind and regional fire detection for every capital
Federation	One country, one standard	Each nation's limits, agencies and languages on one shared engine

Every alert is backed by independent evidence



Promotion tiers, strongest first



Stations and satellites also raise hotspots on their own: an automatic scan checks every city every 30 minutes, with no photo needed.

Report in seconds, in your own language

vayusetu · /report

PUBLIC INTAKE

Report pollution hotspots.

Upload a photo, tag what you see, and share location context. VayuSetu validates the report with AI, local sensors, and satellite signals before alerting municipal responders.

01 Upload photo

Open camera or choose image

Smoke, dust, flame, or visible haze works best.

02 What are you seeing?

Garbage fire
Smoke or flames near waste piles

Traffic smog
Dense fumes at a road junction

Construction dust
Dust plume from digging or debris

Industrial emission
Stack smoke, odor, or chemical haze

03 Location

Reporting in Jakarta, Indonesia. Search, detect or drop a pin; a report is filed under the city that contains its pin.

Location

Detect my location Use map picker

Search for an area...

Use GPS, enter details manually, or open the map picker. No location set

What happens next

- Photo is classified for emission.
- Location is matched with risk zones.
- Verified hotspots are posted on municipal dashboard.

Vayu Setu

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Photo first

Gemini checks the image before anyone sees it on the map.



Voice notes

Transcribed in the city's language: Hindi, Chinese, Russian, Portuguese, Arabic, Indonesian and more.



City-aware location

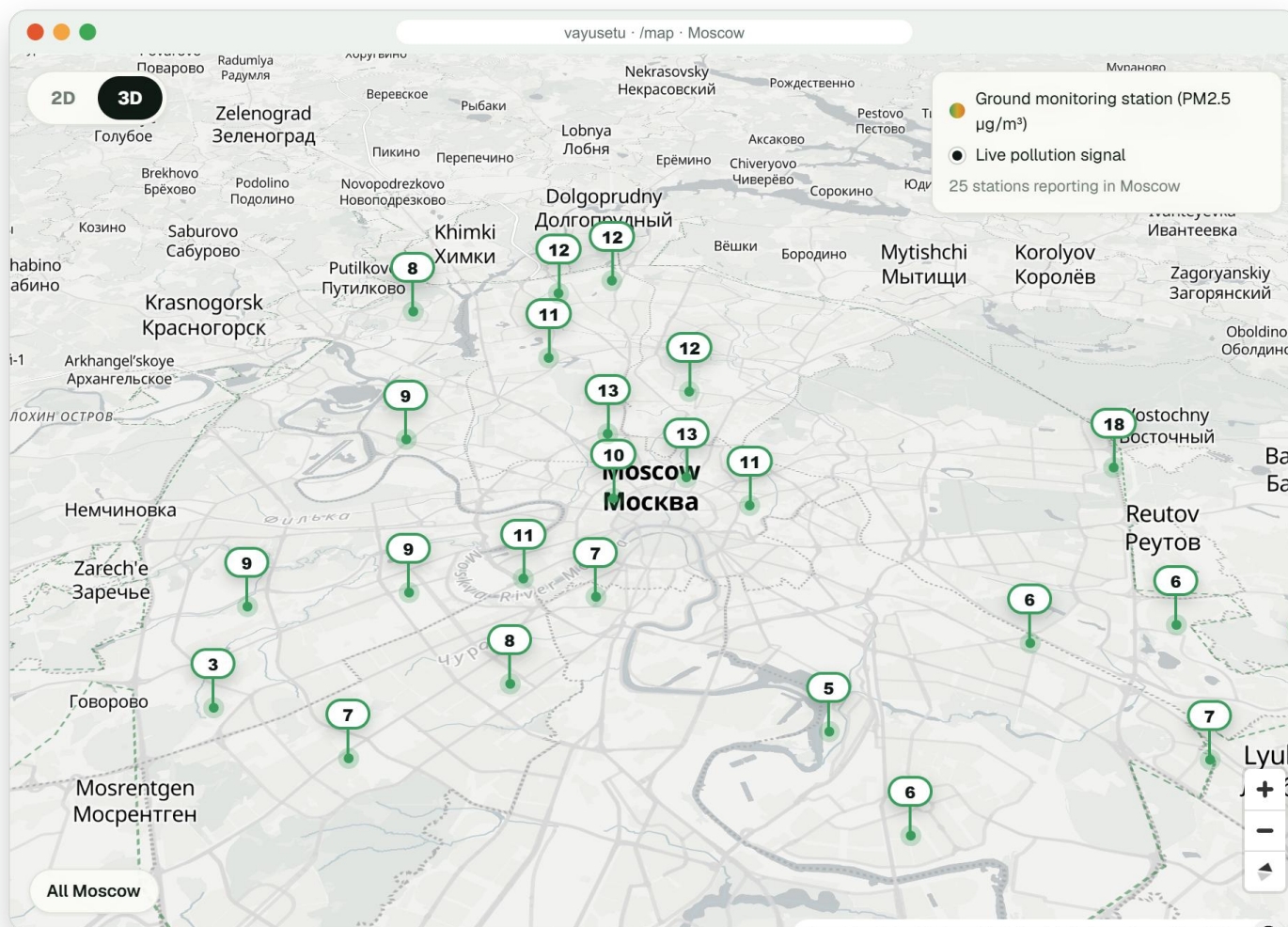
Search, GPS or a pin, scoped to the selected capital.



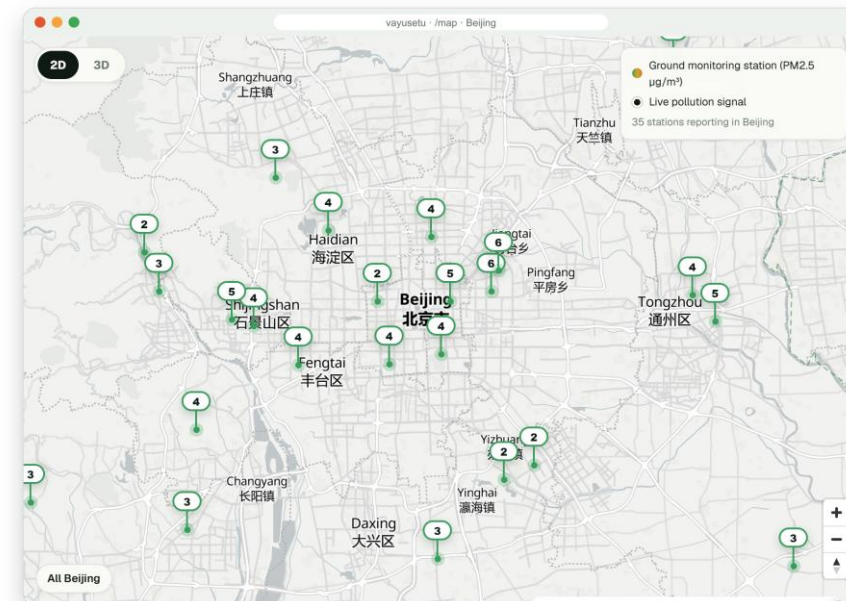
Closed loop

Live status page for every report; reporters confirm or dispute the fix.

Every station, hotspot and fire on a live 3D map



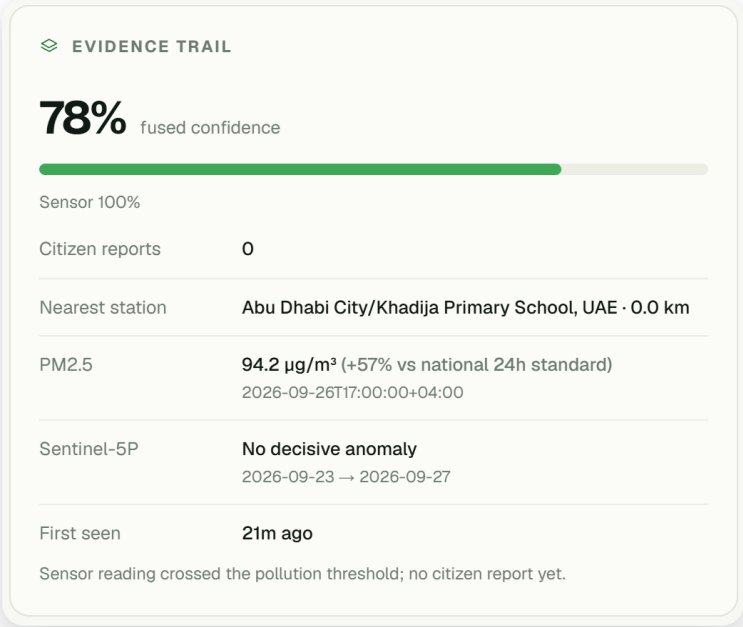
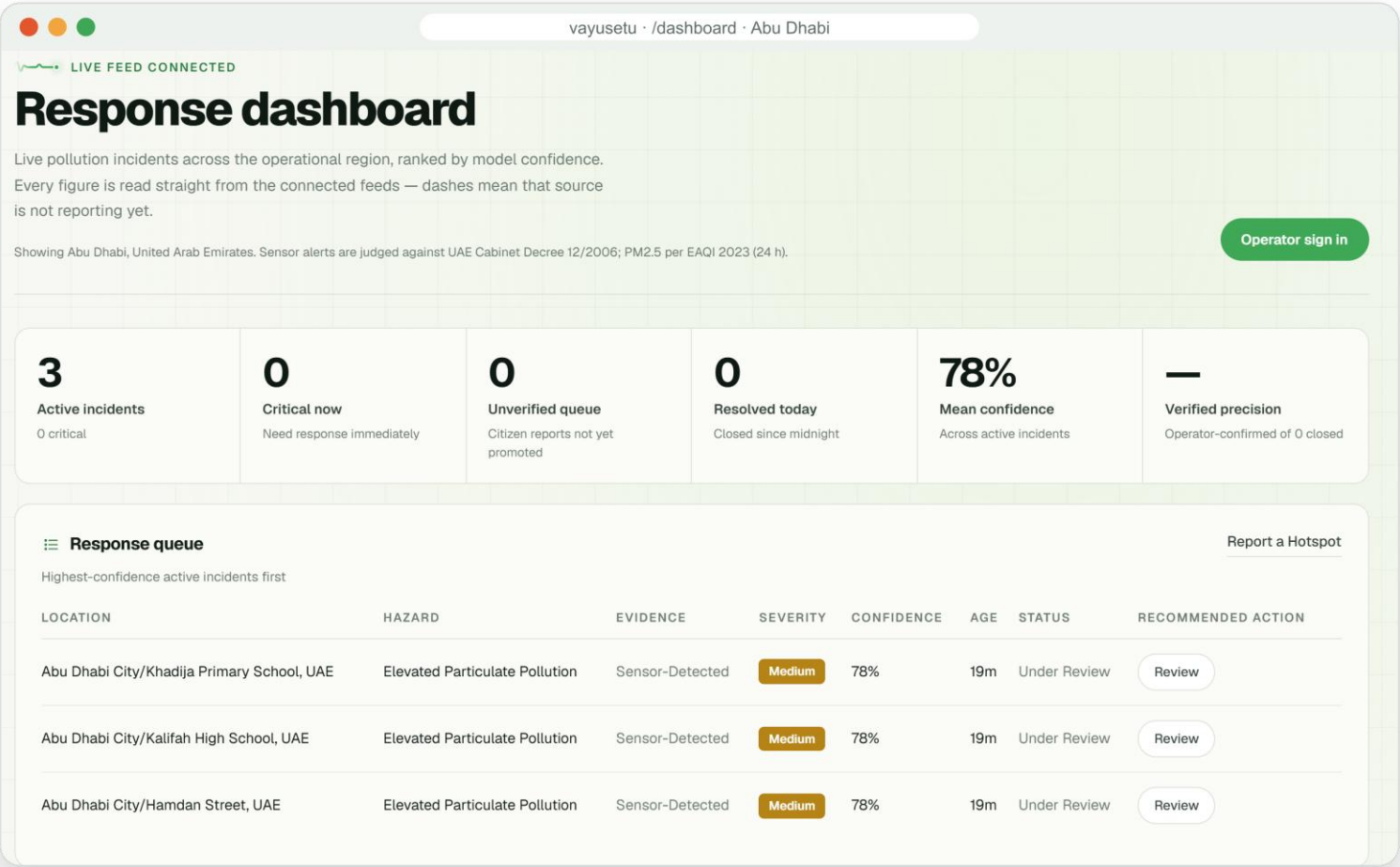
Moscow, 3D view: 25 official stations reporting live



Beijing: 35 live stations

- Stations drawn at their published coordinates, coloured by PM2.5
- OpenStreetMap buildings in 3D; one-tap switch to 2D
- Real city boundary, NASA FIRMS fire layer and live hotspot feed

Operators see what to fix first, and why



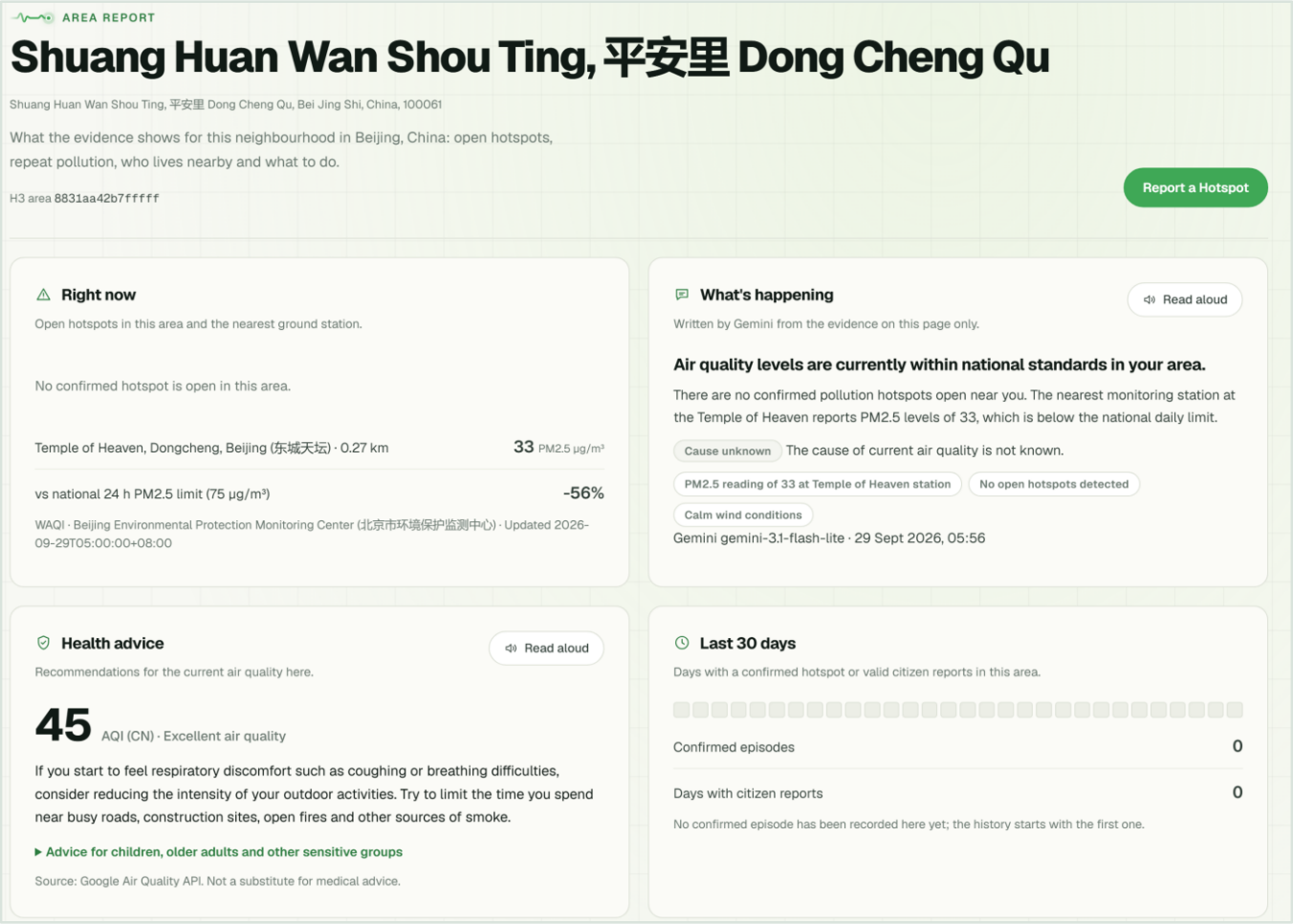
Ranked queue Ordered by evidence tier and fused confidence.

Traceable evidence Station reading vs national limit, satellite window, citizen reports.

One-click action Dispatch, resolve or mark false positive; reporters are notified.

Abu Dhabi, live: three particulate hotspots detected automatically from official EAD stations, judged against the UAE limit.

Every neighbourhood gets its own page



Right now

Open hotspots in the area and the nearest station against the national limit.



Gemini explains it

A plain-language summary built only from the evidence on the page, with read-aloud.



Health advice

Google Air Quality guidance for everyone and six sensitive groups.



Last 30 days

Confirmed episodes and report days; repeat hotspots are flagged.

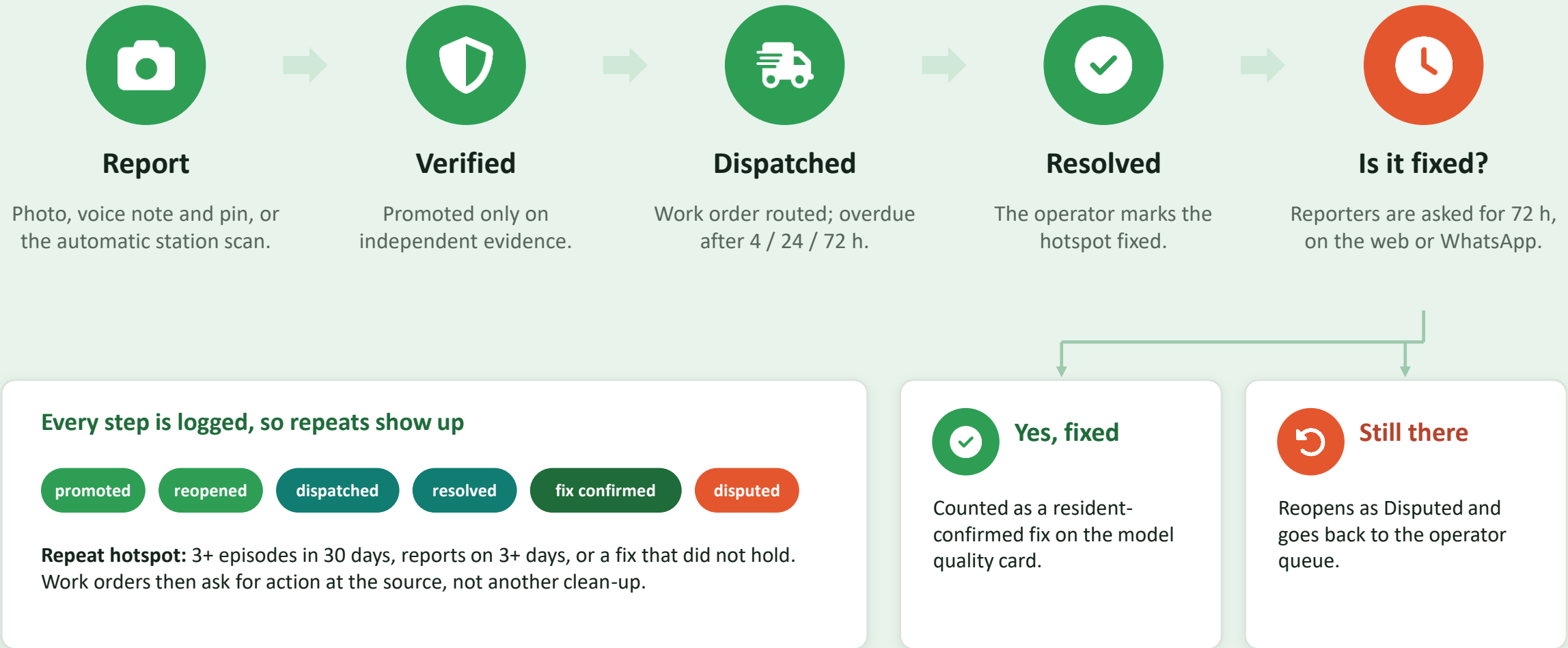


Who's affected

WorldPop residents within 1 km, plus nearby schools and hospitals.

Beijing, live: Temple of Heaven area. Nearest station 0.27 km away; ≈71,838 residents within 1 km (WorldPop 2020).

A fix counts only when residents confirm it



AI that shows its evidence and looks ahead



Gemini

- Vision classifier for smoke, dust, fire and haze, with authenticity checks
- Operations copilot using function calls over live incidents, stations, wind, fires and forecasts
- Work orders in English and the local language, addressed to the responsible agency



Forecasting

- 24-hour PM2.5 outlook for each zone from live station history in BigQuery
- BigQuery ML ARIMA_PLUS model per station, backtested against persistence
- Google Air Quality cross-check in each country's own AQI



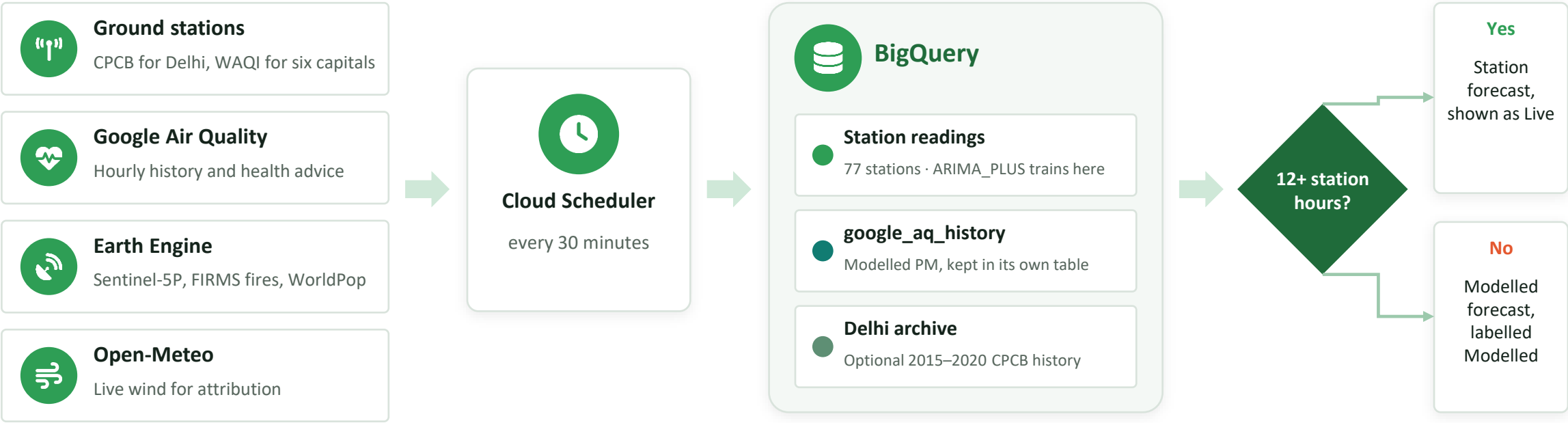
Cross-border attribution

- Live wind plus NASA FIRMS fires across each capital's upwind region
- Punjab–Haryana for Delhi, Hebei for Beijing, the Cerrado for Brasília
- Ranks likely sources: landfills, industrial areas, active fires, open incidents



Grounded, not generated: every copilot answer lists the live tools it called, and every work order cites the evidence behind it.

From raw feeds to a labelled 24-hour outlook



7 / 7

capitals with a 24-hour forecast

77

ground stations streaming into
BigQuery

50

forecast zones with a week of
modelled history

9,191

modelled hourly rows stored

BY THE NUMBERS

What the live system is doing today

7

BRICS capitals, each with its own limits, languages and forecast

77

ground stations streaming into BigQuery every 30 minutes

46 of 48

hotspots raised automatically from stations and satellites, no photo needed

≈71,838

residents within 1 km of Beijing's Temple of Heaven, from WorldPop

13

interface languages, with read-aloud where the device has a voice

72 h

window for residents to confirm or dispute a fix

3

signals behind a repeat-hotspot flag: episodes, report days, failed fixes

43

automated tests, plus a live end-to-end check of the fix-check

Each nation keeps its own rules and language



New Delhi

India

PM2.5 limit 60 $\mu\text{g}/\text{m}^3$

Language Hindi

Stations CPCB / DPCC



Beijing

China

PM2.5 limit 75 $\mu\text{g}/\text{m}^3$

Language Chinese

Stations National network



Moscow

Russia

PM2.5 limit 35 $\mu\text{g}/\text{m}^3$

Language Russian

Stations Mosecomonitoring



Pretoria

South Africa

PM2.5 limit 40 $\mu\text{g}/\text{m}^3$

Language English

Stations SAAQIS



Abu Dhabi

UAE

PM2.5 limit 60 $\mu\text{g}/\text{m}^3$

Language Arabic

Stations EAD



Jakarta

Indonesia

PM2.5 limit 55 $\mu\text{g}/\text{m}^3$

Language Indonesian

Stations KLHK



Brasília

Brazil

PM2.5 limit 50 $\mu\text{g}/\text{m}^3$

Language Portuguese

Stations DF network

A new capital is one configuration entry

Boundary, station network, legal limits, fire region, agencies, languages and time zone. No code changes.

VAYUSETU

Clean air, verified street by street, across BRICS.

Built on Google Cloud

Gemini

Earth Engine

BigQuery ML

Firestore & Auth

App Hosting

Maps & Places

Air Quality API

Speech-to-Text

Cloud Scheduler

Open data

NASA FIRMS

Copernicus Sentinel-5P

WAQI

CPCB

OpenStreetMap

Open-Meteo

Citizens

report in their language and see what happened

Operators

act on ranked, evidence-backed hotspots

Nations

share one engine and keep their own rules

Thank you